

J2EE Architecture Overview

Theory:

Introduction to J2EE and Multi-Tier Architecture:

J2EE (Java 2 Platform, Enterprise Edition), now called Jakarta EE, is a platform used to develop enterprise applications. It follows a multi-tier architecture, where the application is divided into different layers. This makes the application easy to manage, scalable, and maintainable.

The main layers are:

1. **Presentation Layer:**
This layer handles user interaction. It displays data to the user and collects input. Technologies used include Servlets and JSP. It runs inside a web container.
2. **Business Logic Layer:**
This layer contains the main logic of the application. It processes user requests and applies business rules. It runs on an application server.
3. **Data Layer:**
This layer is responsible for storing and retrieving data. It interacts with the database using technologies like JDBC.

Role of Components:

1. **Web Container:**
It manages web components like Servlets and JSP. It handles HTTP requests, responses, and session management.
2. **Application Server:**
It provides services like transaction management, security, and resource management. It runs the business logic.
3. **Database Server:**
It stores and manages the application data.